In [ ]:
```
EXPERIMENT 4 – DATA MANIPULATION WITH PANDAS
PIN NO: 241UDS0109

AIM:
To perform data manipulation operations using Pandas: importing a dataset, setting an index, selecting rows and columns, creating
calculated columns, filtering, sorting, updating values, and renaming columns.
```

In [ ]:
```
## Step 1 – Install and Import Pandas Library
If Pandas is not installed, run `%pip install pandas` in a separate code cell.
```

In [1]:
```
import pandas as pd
```

In [ ]:
```
## Step 2 – Importing the Dataset
```

In [2]:
```
df_241UDS0109 = pd.read_csv("Experiment_4_241UDS0109_Urban_Garden_Supplies.csv")
df_241UDS0109.head(5)
```

Out[2]: Columns 1–8 of 11

	Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
0	G2001	Ceramic Planter	Garden	Vizag	34	420	5
1	G2002	Pruning Shears	Tools	Rajahmundry	61	580	8
2	G2003	Organic Compost	Soil Care	Vijayawada	48	260	10
3	G2004	Watering Can	Garden	Hyderabad	27	350	6
4	G2005	Seed Starter Tray	Planting	Pune	73	140	12

Columns 9–11 of 11 · first column repeated

	Customer_Rating	Stock_Available	Payment_Method
0	4.4	22	UPI
1	4.6	17	Card
2	4.3	40	Cash
3	4.1	15	UPI
4	4.5	55	Card

In [ ]:
```
## Step 3 – Set Index for the DataFrame
```

In [3]:
```
df_241UDS0109 = df_241UDS0109.set_index("Product_ID")
df_241UDS0109.head()
```

Out[3]: Columns 1–8 of 10

Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
G2001	Ceramic Planter	Garden	Vizag	34	420	5	4.4
G2002	Pruning Shears	Tools	Rajahmundry	61	580	8	4.6
G2003	Organic Compost	Soil Care	Vijayawada	48	260	10	4.3
G2004	Watering Can	Garden	Hyderabad	27	350	6	4.1
G2005	Seed Starter Tray	Planting	Pune	73	140	12	4.5

Columns 9–10 of 10 · first column repeated

Product_ID	Stock_Available	Payment_Method
G2001	22	UPI
G2002	17	Card
G2003	40	Cash
G2004	15	UPI
G2005	55	Card

In [ ]:
```
## Step 4 – Info of the DataFrame
```

In [4]:
```
df_241UDS0109.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 30 entries, G2001 to G2030
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Product_Name      30 non-null     object
 1   Category          30 non-null     object
 2   City              30 non-null     object
 3   Units_Sold        30 non-null     int64
 4   Unit_Price        30 non-null     int64
 5   Discount_Percent  30 non-null     int64
 6   Customer_Rating   30 non-null     float64
 7   Stock_Available   30 non-null     int64
 8   Payment_Method    30 non-null     object
dtypes: float64(1), int64(4), object(4)
memory usage: 2.3+ KB
```

In [ ]:
```
## Step 5 — Columns & Rows Selection
### Selecting a Single Column
```

In [5]:
```
df_241UDS0109["Product_Name"]
```

Out[5]:
```
Product_ID
G2001        Ceramic Planter
G2002         Pruning Shears
G2003        Organic Compost
G2004           Watering Can
G2005      Seed Starter Tray
G2006          Garden Gloves
G2007         Hose Connector
G2008             Neem Spray
G2009         Raised Bed Kit
G2010           Plant Labels
G2011             Hand Trowel
G2012        Coco Peat Block
G2013    Drip Irrigation Set
G2014          Seed Mix Pack
G2015       Fertilizer Scoop
G2016            Bird Feeder
G2017           Plant Mister
G2018           Slug Barrier
G2019           Garden Twine
G2020            Potting Mix
G2021      Solar Path Light
G2022              Leaf Rake
G2023        Hanging Basket
G2024    Plant Support Clips
G2025           Rooting Gel
G2026    Compost Thermometer
G2027        Mini Greenhouse
G2028         Garden Kneeler
G2029            Pruning Saw
G2030       Plant Food Drops
Name: Product_Name, dtype: object
```

In [ ]:
```
## Step 6 — Selecting Multiple Columns
```

In [6]:
```
df_241UDS0109[["Product_Name", "Category", "City"]].head(3)
```

Out[6]:

Product_ID	Product_Name	Category	City
G2001	Ceramic Planter	Garden	Vizag
G2002	Pruning Shears	Tools	Rajahmundry
G2003	Organic Compost	Soil Care	Vijayawada

In [ ]:
```
## Step 7 — Select Row Using iloc
```

In [7]:
```
df_241UDS0109.iloc[1]
```

Out[7]:
```
Product_Name        Pruning Shears
Category                     Tools
City                   Rajahmundry
Units_Sold                      61
Unit_Price                     580
Discount_Percent                 8
Customer_Rating                4.6
Stock_Available                 17
Payment_Method                Card
Name: G2002, dtype: object
```

In [ ]:
```
## Step 8 — Select Row Using loc
Select the last product by its product ID.
```

In [8]:
```
df_241UDS0109.loc["G2030"]
```

Out[8]:
```
Product_Name        Plant Food Drops
Category                  Plant Care
City                            Pune
Units_Sold                        59
Unit_Price                       240
Discount_Percent                   8
Customer_Rating                  4.4
Stock_Available                   31
Payment_Method                  Card
Name: G2030, dtype: object
```

In [ ]:
```
## Step 9 — Manipulating Data Based on Columns
Display the first two rows.
```

In [9]:
```
df_241UDS0109.head(2)
```

Out[9]: Columns 1–8 of 10

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
G2001	Ceramic Planter	Garden	Vizag	34	420	5	4.4
G2002	Pruning Shears	Tools	Rajahmundry	61	580	8	4.6

Columns 9–10 of 10 · first column repeated

	Stock_Available	Payment_Method
Product_ID		
G2001	22	UPI
G2002	17	Card

In [ ]:
```
## Step 10 — Creating Total Sales Column
```

In [10]:
```
df_241UDS0109["Total_Sales_241UDS0109"] = df_241UDS0109["Units_Sold"] * df_241UDS0109["Unit_Price"]
df_241UDS0109.head(2)
```

Out[10]: Columns 1–8 of 11

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
G2001	Ceramic Planter	Garden	Vizag	34	420	5	4.4
G2002	Pruning Shears	Tools	Rajahmundry	61	580	8	4.6

Columns 9–11 of 11 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0109
Product_ID			
G2001	22	UPI	14280
G2002	17	Card	35380

In [ ]:
```
## Step 11 — Creating Discount Amount Column
```

In [11]:
```
df_241UDS0109["Discount_Amount_241UDS0109"] = df_241UDS0109["Total_Sales_241UDS0109"] * df_241UDS0109["Discount_Percent"] / 100
df_241UDS0109.head(2)
```

Out[11]: Columns 1–8 of 12

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
G2001	Ceramic Planter	Garden	Vizag	34	420	5	4.4
G2002	Pruning Shears	Tools	Rajahmundry	61	580	8	4.6

Columns 9–12 of 12 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109
Product_ID				
G2001	22	UPI	14280	714.0
G2002	17	Card	35380	2830.4

In [ ]:
```
## Step 12 — Creating Net Sales Column
```

In [12]:
```
df_241UDS0109["Net_Sales_241UDS0109"] = df_241UDS0109["Total_Sales_241UDS0109"] - df_241UDS0109["Discount_Amount_241UDS0109"]
df_241UDS0109.head(2)
```

Out[12]: Columns 1–8 of 13

Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
G2001	Ceramic Planter	Garden	Vizag	34	420	5	4.4
G2002	Pruning Shears	Tools	Rajahmundry	61	580	8	4.6

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Net_Sales_241UDS0109
G2001	22	UPI	14280	714.0	13566.0
G2002	17	Card	35380	2830.4	32549.6

In [ ]:
```
## Step 13 — Round Calculated Values
```

In [13]:
```
df_241UDS0109["Discount_Amount_241UDS0109"] = df_241UDS0109["Discount_Amount_241UDS0109"].round(2)
df_241UDS0109["Net_Sales_241UDS0109"] = df_241UDS0109["Net_Sales_241UDS0109"].round(2)
df_241UDS0109.head(2)
```

Out[13]: Columns 1–8 of 13

Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
G2001	Ceramic Planter	Garden	Vizag	34	420	5	4.4
G2002	Pruning Shears	Tools	Rajahmundry	61	580	8	4.6

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Net_Sales_241UDS0109
G2001	22	UPI	14280	714.0	13566.0
G2002	17	Card	35380	2830.4	32549.6

In [ ]:
```
## Step 14 — Conditional Filtering
Display products with more than 50 units sold.
```

In [14]:
```
df_241UDS0109[df_241UDS0109["Units_Sold"] > 50].head(2)
```

Out[14]: Columns 1–8 of 13

Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
G2002	Pruning Shears	Tools	Rajahmundry	61	580	8	4.6
G2005	Seed Starter Tray	Planting	Pune	73	140	12	4.5

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Net_Sales_241UDS0109
G2002	17	Card	35380	2830.4	32549.6
G2005	55	Card	10220	1226.4	8993.6

In [ ]:
```
## Step 15 — Products with High Sales AND Low Stock
Find products with total sales above 15,000 and stock below 25.
```

In [15]:
```
df_241UDS0109[(df_241UDS0109["Total_Sales_241UDS0109"] > 15000) & (df_241UDS0109["Stock_Available"] < 25)].head(3)
```

Out[15]: Columns 1–8 of 13

Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
G2002	Pruning Shears	Tools	Rajahmundry	61	580	8	4.6
G2009	Raised Bed Kit	Garden	Vizag	22	1850	11	4.8
G2013	Drip Irrigation Set	Watering	Chennai	31	1250	13	4.7

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Net_Sales_241UDS0109
G2002	17	Card	35380	2830.4	32549.6
G2009	12	UPI	40700	4477.0	36223.0
G2013	14	Card	38750	5037.5	33712.5

In [ ]:
```
## Step 16 — Products from Vizag OR Hyderabad
```

In [16]:
```
df_241UDS0109[(df_241UDS0109["City"] == "Vizag") | (df_241UDS0109["City"] == "Hyderabad")].head()
```

Out[16]: Columns 1–8 of 13

Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
G2001	Ceramic Planter	Garden	Vizag	34	420	5	4.4
G2004	Watering Can	Garden	Hyderabad	27	350	6	4.1
G2009	Raised Bed Kit	Garden	Vizag	22	1850	11	4.8
G2011	Hand Trowel	Tools	Hyderabad	57	290	6	4.4
G2017	Plant Mister	Watering	Vizag	43	270	9	4.5

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Net_Sales_241UDS0109
G2001	22	UPI	14280	714.0	13566.0
G2004	15	UPI	9450	567.0	8883.0
G2009	12	UPI	40700	4477.0	36223.0
G2011	33	Card	16530	991.8	15538.2
G2017	24	UPI	11610	1044.9	10565.1

In [ ]:
```
## Step 17 — Products with Rating ≥ 4.5 AND Stock < 30
```

In [17]:
```
df_241UDS0109[(df_241UDS0109["Customer_Rating"] ⩾ 4.5) & (df_241UDS0109["Stock_Available"] < 30)]
```

Out[17]: Columns 1–8 of 13

Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
G2002	Pruning Shears	Tools	Rajahmundry	61	580	8	4.6
G2008	Neem Spray	Plant Care	Delhi	45	310	9	4.6
G2009	Raised Bed Kit	Garden	Vizag	22	1850	11	4.8
G2013	Drip Irrigation Set	Watering	Chennai	31	1250	13	4.7
G2016	Bird Feeder	Outdoor	Mumbai	36	740	7	4.6
G2017	Plant Mister	Watering	Vizag	43	270	9	4.5
G2020	Potting Mix	Soil Care	Hyderabad	53	320	10	4.7
G2023	Hanging Basket	Garden	Delhi	32	510	9	4.6
G2025	Rooting Gel	Plant Care	Rajahmundry	28	640	14	4.8
G2026	Compost Thermometer	Soil Care	Vizag	17	890	7	4.5
G2027	Mini Greenhouse	Garden	Bengaluru	24	1450	10	4.7
G2029	Pruning Saw	Tools	Hyderabad	38	670	11	4.6

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Net_Sales_241UDS0109
G2002	17	Card	35380	2830.4	32549.6
G2008	28	Card	13950	1255.5	12694.5
G2009	12	UPI	40700	4477.0	36223.0
G2013	14	Card	38750	5037.5	33712.5
G2016	16	Card	26640	1864.8	24775.2
G2017	24	UPI	11610	1044.9	10565.1
G2020	26	UPI	16960	1696.0	15264.0
G2023	17	UPI	16320	1468.8	14851.2
G2025	13	Card	17920	2508.8	15411.2
G2026	9	UPI	15130	1059.1	14070.9
G2027	15	Card	34800	3480.0	31320.0
G2029	19	UPI	25460	2800.6	22659.4

In [ ]:
```
## Step 18 — Filtering Multiple Cities Using .isin()
```

In [18]:
```
df_241UDS0109[df_241UDS0109["City"].isin(["Vizag", "Bengaluru", "Pune"])].head()
```

Out[18]: Columns 1–8 of 13

Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
G2001	Ceramic Planter	Garden	Vizag	34	420	5	4.4
G2005	Seed Starter Tray	Planting	Pune	73	140	12	4.5
G2007	Hose Connector	Tools	Bengaluru	39	230	7	4.2
G2009	Raised Bed Kit	Garden	Vizag	22	1850	11	4.8
G2012	Coco Peat Block	Soil Care	Pune	66	175	8	4.5

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Net_Sales_241UDS0109
G2001	22	UPI	14280	714.0	13566.0
G2005	55	Card	10220	1226.4	8993.6
G2007	19	UPI	8970	627.9	8342.1
G2009	12	UPI	40700	4477.0	36223.0
G2012	44	UPI	11550	924.0	10626.0

In [ ]:
```
## Step 19 — Filtering Multiple Categories Using .isin()
```

In [19]:
```
df_241UDS0109[df_241UDS0109["Category"].isin(["Garden", "Tools"])].head()
```

Out[19]: Columns 1–8 of 13

Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
G2001	Ceramic Planter	Garden	Vizag	34	420	5	4.4
G2002	Pruning Shears	Tools	Rajahmundry	61	580	8	4.6
G2004	Watering Can	Garden	Hyderabad	27	350	6	4.1
G2007	Hose Connector	Tools	Bengaluru	39	230	7	4.2
G2009	Raised Bed Kit	Garden	Vizag	22	1850	11	4.8

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Net_Sales_241UDS0109
G2001	22	UPI	14280	714.0	13566.0
G2002	17	Card	35380	2830.4	32549.6
G2004	15	UPI	9450	567.0	8883.0
G2007	19	UPI	8970	627.9	8342.1
G2009	12	UPI	40700	4477.0	36223.0

In [ ]:
```
## Step 20 — Filtering Multiple Payment Methods Using .isin()
```

In [20]:
```
df_241UDS0109[df_241UDS0109["Payment_Method"].isin(["UPI", "Card"])].head()
```

Out[20]: Columns 1–8 of 13

Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
G2001	Ceramic Planter	Garden	Vizag	34	420	5	4.4
G2002	Pruning Shears	Tools	Rajahmundry	61	580	8	4.6
G2004	Watering Can	Garden	Hyderabad	27	350	6	4.1
G2005	Seed Starter Tray	Planting	Pune	73	140	12	4.5
G2007	Hose Connector	Tools	Bengaluru	39	230	7	4.2

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Net_Sales_241UDS0109
G2001	22	UPI	14280	714.0	13566.0
G2002	17	Card	35380	2830.4	32549.6
G2004	15	UPI	9450	567.0	8883.0
G2005	55	Card	10220	1226.4	8993.6
G2007	19	UPI	8970	627.9	8342.1

In [ ]:
```
## Step 21 — Sorting Table
Sort by Units Sold in descending order.
```

In [21]:
```
df_241UDS0109.sort_values("Units_Sold", ascending=False).head()
```

Out[21]: Columns 1–8 of 13

Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
G2006	Garden Gloves	Safety	Chennai	96	180	15	4.7
G2014	Seed Mix Pack	Planting	Rajahmundry	92	120	10	4.3
G2024	Plant Support Clips	Planting	Mumbai	88	75	5	4.1
G2010	Plant Labels	Planting	Mumbai	84	95	5	4.0
G2019	Garden Twine	Planting	Vijayawada	78	110	6	4.0

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Net_Sales_241UDS0109
G2006	62	Cash	17280	2592.0	14688.0
G2014	58	UPI	11040	1104.0	9936.0
G2024	65	Cash	6600	330.0	6270.0
G2010	70	Cash	7980	399.0	7581.0
G2019	49	Cash	8580	514.8	8065.2

In [ ]:
```
## Step 22 — Sort by Unit Price — Ascending
```

In [22]:
```
df_241UDS0109.sort_values("Unit_Price", ascending=True).head()
```

Out[22]: Columns 1–8 of 13

Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
G2024	Plant Support Clips	Planting	Mumbai	88	75	5	4.1
G2010	Plant Labels	Planting	Mumbai	84	95	5	4.0
G2019	Garden Twine	Planting	Vijayawada	78	110	6	4.0
G2014	Seed Mix Pack	Planting	Rajahmundry	92	120	10	4.3
G2005	Seed Starter Tray	Planting	Pune	73	140	12	4.5

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Net_Sales_241UDS0109
G2024	65	Cash	6600	330.0	6270.0
G2010	70	Cash	7980	399.0	7581.0
G2019	49	Cash	8580	514.8	8065.2
G2014	58	UPI	11040	1104.0	9936.0
G2005	55	Card	10220	1226.4	8993.6

In [ ]:
```
## Step 23 — Sorting Using Multiple Columns
Sort by Category ascending and Units Sold descending.
```

In [23]:
```
df_241UDS0109.sort_values(["Category", "Units_Sold"], ascending=[True, False]).head()
```

Out[23]: Columns 1–8 of 13

Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
G2001	Ceramic Planter	Garden	Vizag	34	420	5	4.4
G2023	Hanging Basket	Garden	Delhi	32	510	9	4.6
G2004	Watering Can	Garden	Hyderabad	27	350	6	4.1
G2027	Mini Greenhouse	Garden	Bengaluru	24	1450	10	4.7
G2009	Raised Bed Kit	Garden	Vizag	22	1850	11	4.8

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Net_Sales_241UDS0109
G2001	22	UPI	14280	714.0	13566.0
G2023	17	UPI	16320	1468.8	14851.2
G2004	15	UPI	9450	567.0	8883.0
G2027	15	Card	34800	3480.0	31320.0
G2009	12	UPI	40700	4477.0	36223.0

In [ ]:
```
## Step 24 — Updating Data in the Table
Replace the city name Vizag with VZG.
```

In [24]:
```
df_241UDS0109["City"] = df_241UDS0109["City"].replace(["Vizag"], ["VZG"])
df_241UDS0109[df_241UDS0109["City"] == "VZG"].head(2)
```

Out[24]: Columns 1–8 of 13

Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
G2001	Ceramic Planter	Garden	VZG	34	420	5	4.4
G2009	Raised Bed Kit	Garden	VZG	22	1850	11	4.8

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Net_Sales_241UDS0109
G2001	22	UPI	14280	714.0	13566.0
G2009	12	UPI	40700	4477.0	36223.0

In [ ]:
```
## Step 25 — Updating a Specific Product Value
Update the unit price for product G2001.
```

In [25]:
```
df_241UDS0109.loc["G2001", "Unit_Price"] = 450
print(df_241UDS0109.loc["G2001", "Unit_Price"])
```

```
450
```

In [ ]:
```
## Step 26 — Changing Column Names
```

In [26]:
```
df_241UDS0109 = df_241UDS0109.rename(columns={
    "Product_Name": "Item_Name",
    "City": "Location",
    "Customer_Rating": "Rating",
    "Net_Sales_241UDS0109": "Revenue"
})
df_241UDS0109.head(1)
```

Out[26]: Columns 1–8 of 13

Product_ID	Item_Name	Category	Location	Units_Sold	Unit_Price	Discount_Percent	Rating
G2001	Ceramic Planter	Garden	VZG	34	450	5	4.4

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Revenue
G2001	22	UPI	14280	714.0	13566.0

In [ ]:
```
## Step 27 — Final DataFrame
```

In [27]:
```
df_241UDS0109.head()
```

Out[27]: Columns 1–8 of 13

Product_ID	Item_Name	Category	Location	Units_Sold	Unit_Price	Discount_Percent	Rating
G2001	Ceramic Planter	Garden	VZG	34	450	5	4.4
G2002	Pruning Shears	Tools	Rajahmundry	61	580	8	4.6
G2003	Organic Compost	Soil Care	Vijayawada	48	260	10	4.3
G2004	Watering Can	Garden	Hyderabad	27	350	6	4.1
G2005	Seed Starter Tray	Planting	Pune	73	140	12	4.5

Columns 9–13 of 13 · first column repeated

Product_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0109	Discount_Amount_241UDS0109	Revenue
G2001	22	UPI	14280	714.0	13566.0
G2002	17	Card	35380	2830.4	32549.6
G2003	40	Cash	12480	1248.0	11232.0
G2004	15	UPI	9450	567.0	8883.0
G2005	55	Card	10220	1226.4	8993.6

In [ ]:
```
## Step 28 — Final Dataset Information
```

In [28]:
```
df_241UDS0109.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 30 entries, G2001 to G2030
Data columns (total 12 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Item_Name                   30 non-null     object
 1   Category                    30 non-null     object
 2   Location                    30 non-null     object
 3   Units_Sold                  30 non-null     int64
 4   Unit_Price                  30 non-null     int64
 5   Discount_Percent            30 non-null     int64
 6   Rating                      30 non-null     float64
 7   Stock_Available             30 non-null     int64
 8   Payment_Method              30 non-null     object
 9   Total_Sales_241UDS0109      30 non-null     int64
 10  Discount_Amount_241UDS0109  30 non-null     float64
 11  Revenue                     30 non-null     float64
dtypes: float64(3), int64(5), object(4)
memory usage: 4.1+ KB
```

In [ ]:
```
RESULT:-
Pandas operations were performed on the urban-garden supplies dataset, including indexing, column and row selection, calculated sales
and discount columns, conditional filtering, sorting, updating values, and renaming columns. PIN: 241UDS0109.
```